

SMS 100 – Introduction to Ocean Sciences

Tiago Carrilho Biló

SMS - Physical
Oceanographer specialized
in large scale ocean
circulation.

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Class Format: This is an ***In-Person*** (Face-to-Face) lecture class; classes are held Tuesdays and Thursdays, 9:30–10:45 am, in 101 Neville Hall.

However, synchronous Zoom access will be available if another “COVID-like” situation materializes, for example.

Course Description: SMS 100, *Introduction to Ocean Science*, is a lecture-based class that surveys the broad field of marine science, which includes the four main sub-discipline of oceanography: physical, chemical, biological, and geological oceanography.

There are no prerequisites, although it is assumed that all students will have had the typical high school math and science background required for admission to the university. SMS 100 introduces the many facets of the marine realm, including basic ocean physics, biology, chemistry, geology, and the diverse marine environments and marine life forms we find there. Oceanography, as we will emphasize, is not just about these individual aspects, but about the rich and complex interconnections among them, living and non-living marine resources, and human interactions with the natural environment.

SMS 100 – Fall 2025 - Course Summary

- Introduction to Oceanography
- Course materials will be presented in in-person lectures
 - Weeks 1-4: Marine Geology
 - Weeks 5-8: Marine Chemistry
 - Weeks 9-12: Physical Oceanography
 - Week 12-15: Biological Oceanography
- **Learning Outcome: to make you think about marine environments relevant to you and identify their important components that make them unique**

SMS 100 – Fall 2025 - Course Summary

To achieve this learning outcome it is my hope to answer fundamental questions about the ocean grouped in the following topics:

1. Navigation and early ocean exploration;
2. History of the universe, earth, life, and the ocean;
3. Ocean floor;
4. Water;
5. Water in movement: ocean current and waves;
6. Life in the ocean;
7. Challenges and human interactions.

SMS 100 – Fall 2025 - Course Logistics

Office Hours:

Monday and Wednesday mornings at 10:00 am to 12:00 pm.

Graduate Teaching Assistants:

We have two graduate Teaching Assistants (*a.k.a.* “TAs”) this semester:

Collin Beirne (collin.beirne@maine.edu) is a Marine Science Master’s degree student. He holds a B.Sc. in Marine Science, which he earned here at UMaine.

Ana Clara Mariani (ana.mariani@maine.edu) is a Marine Science Ph.D. student. She holds a B.Sc. in Oceanography and M.Sc. in Physical Oceanography from the University of Sao Paulo, Brazil.

TAs will also hold office ours if you are busy during my office hours

SMS 100 – Fall 2025 - Course Logistics

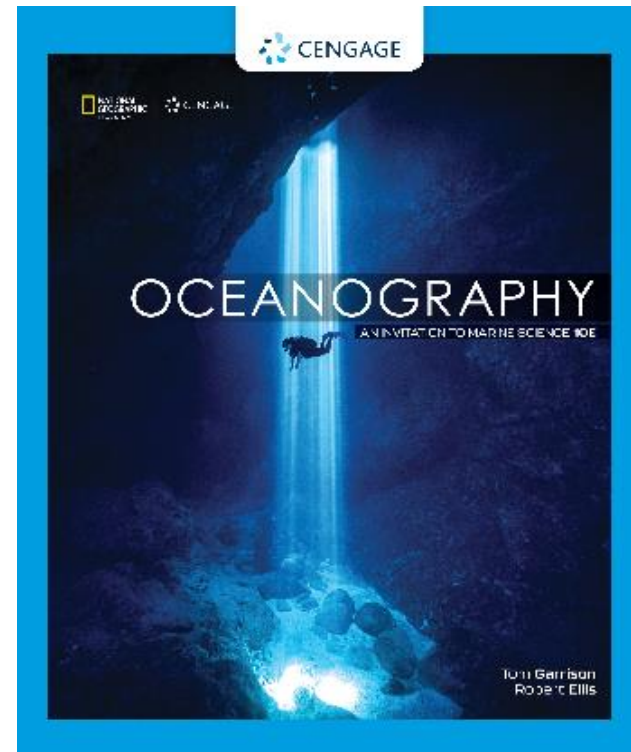
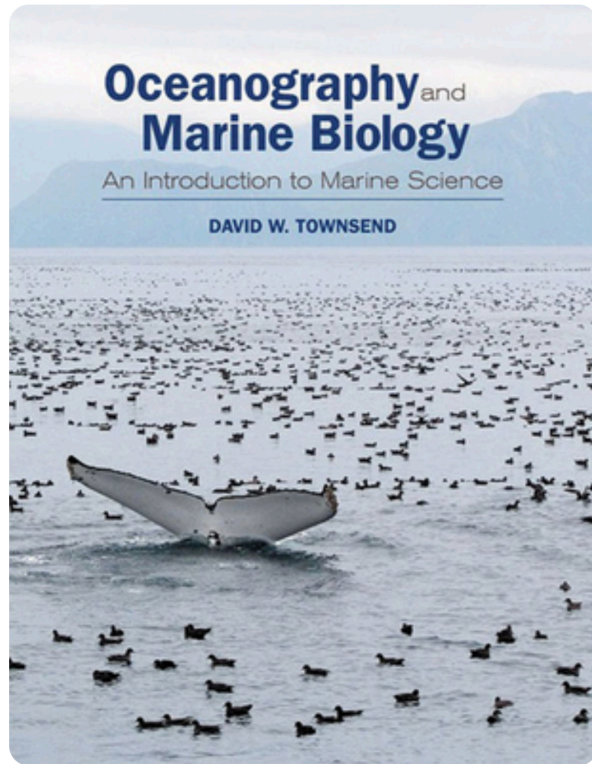
Brightspace:

We will maintain online a ***Brightspace*** LMP (Learning Management Platform) web page for this class, which you already have access to via the **UMaine Campus Portal** (or at: <https://courses.maine.edu/>). Periodic announcements will be posted there, in addition to the following basic course materials:

You will find everything there! From course documents, slides, notes, videos, announcements, etc.

SMS 100 – Fall 2025 - Course Logistics

Textbooks



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Exams and grading:

There will be two (2) exams and four (4) homework assignments. Each exam is worth 100 pts and each homework assignment is worth 50 pts, for a possible semester total of 400 pts; your grade will be based on the percentage of the 400 pts.

<u>Letter Grade</u>	<u>Percentage</u>	<u>Points</u>
A+	93.50 - 100%	374.00 - 400.00
A-	90.00 - 93.49%	360.00 - 373.96
B+	86.67 - 89.99%	346.68 - 359.99
B	83.34 - 86.66%	333.36 - 346.67
B-	80.00 - 83.33%	320.00 - 333.35
C+	76.67 - 79.99%	306.68 - 319.99
C	73.34 - 76.66%	293.36 - 306.67
C-	70.00 - 73.33%	280.00 - 293.35
D+	66.67 - 69.99%	266.68 - 279.99
D	63.34 - 66.66%	253.36 - 266.67
D-	60.00 - 63.33%	240.00 - 253.35
F	<60%	<240

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Makeup Exams:

There are no makeup exams, except under unusual circumstances, in addition to the examples covered in the section on *Missed Class Policy and University Sanctioned Absences*.

If you know ahead of time that missing an exam cannot be avoided, you must provide me with written documentation well beforehand, **preferably by the second week of classes**.

Extra-Credit:

Participation and engagement in class are highly encouraged. I will do my best to reward you if you ask questions and engage in class discussions. Final amount of individual extra-credit will be decided by me at the end of the semester and it can be as high as fifty (50) pts.

Class Attendance:

Class attendance is **not mandatory**. However, exams will include material covered in lecture and class discussions, which **may not be** covered in your textbook.



Lecture 1

1. Importance of ocean sciences;
2. Basic facts about our oceans and oceanography.

Why study the oceans?



Yes, it is important!

Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;



Fishing

Salt production



Tom Garrison



Fish oil

Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;

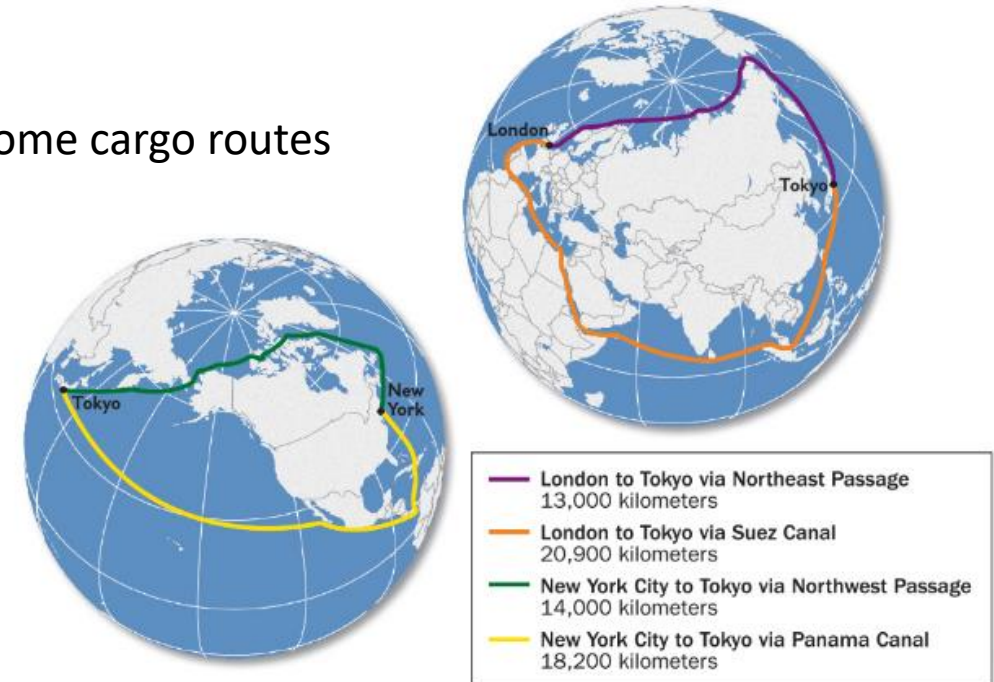
2. Transportation;

Panama Canal lock gates



AP images/Arnulfo Franco

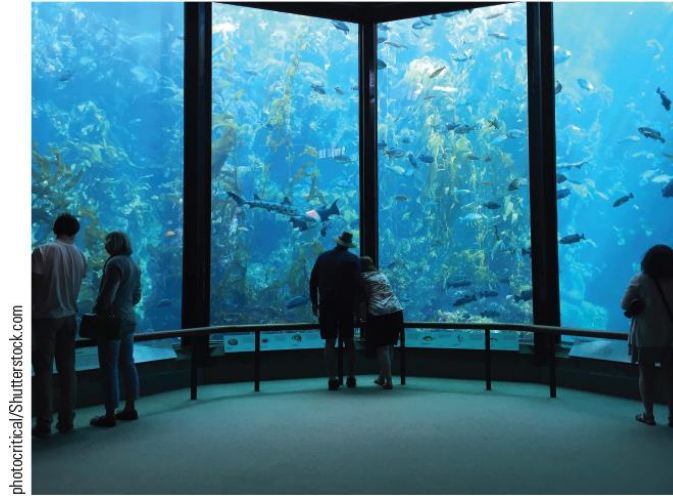
Some cargo routes



Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;



Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;
4. Biodiversity;



Kelp forest views from below the ocean surface. Credit: Andrew b Stowe

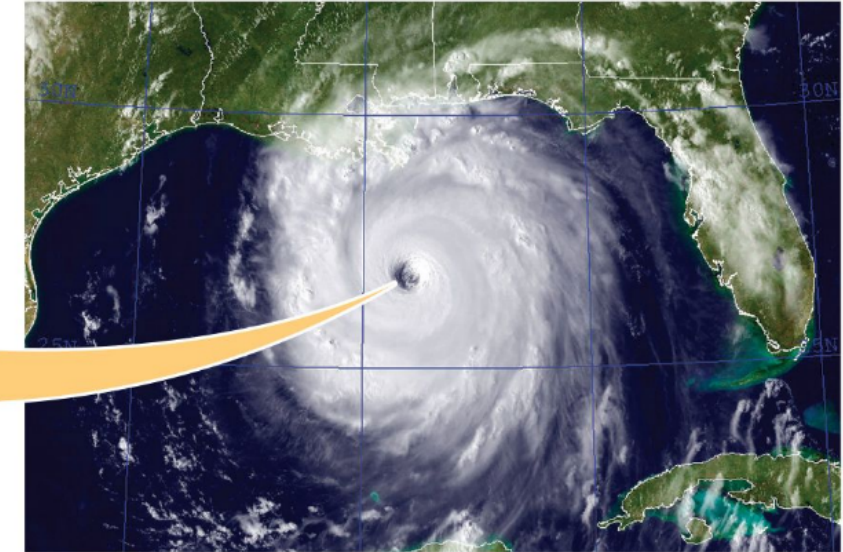
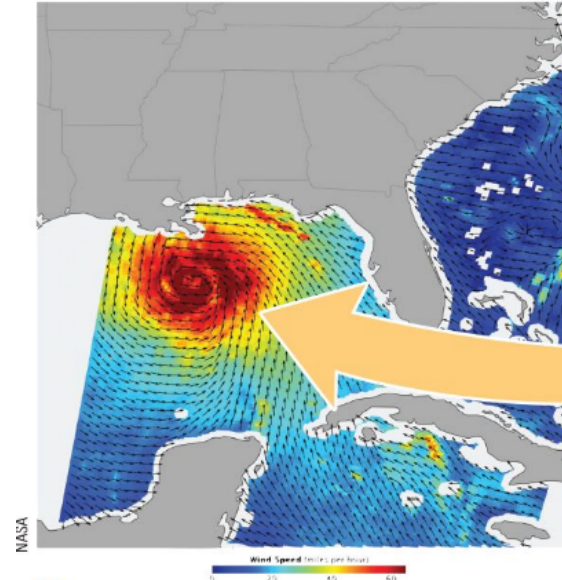
Nearly half of the planet's inhabitants live within 240 kilometers (150 miles) of a coastline (very close to the ocean)

Why study the oceans?

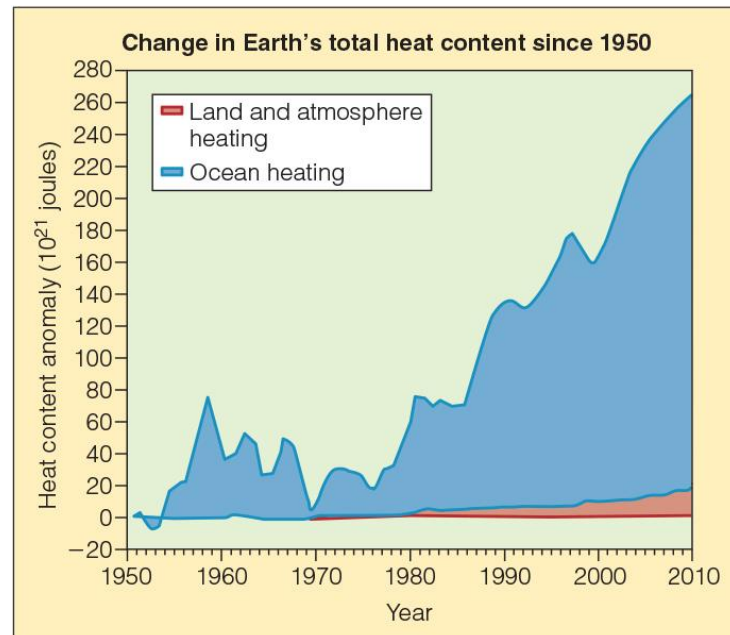
A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;
4. Biodiversity;
5. Import from local weather and conditions to global climate

Hurricane Katrina August 28, 2005



a Katrina as a category 5 storm hours before landfall on 28 August 2005. The storm is relatively compact with strong winds confined to about 500 kilometers (310 miles) from edge to edge. Katrina's overall shape and well-formed eye is easily seen.



Oceans slow down global warming by sequestering a lot of heat from our atmosphere. Consequence
Example: man-forced sea level rise.



Studying the Oceans: Oceanography

Oceanography (or marine science) is the scientific study of the ocean, its associated life-forms, and its bordering lands

- Marine science draws on many disciplines: geology, physics, biology, chemistry, astronomy, atmospheric science, anthropology, ecology, computer science, applied mathematics, and engineering.
- As a marine scientist, we specialize in one area of research, but we must be familiar with related disciplines and appreciate the linkages between them.

Oceanography professionals

- Marine geologists
- Physical oceanographers
- Chemical oceanographers
- Climate specialists
- Marine biologists
- Marine engineers



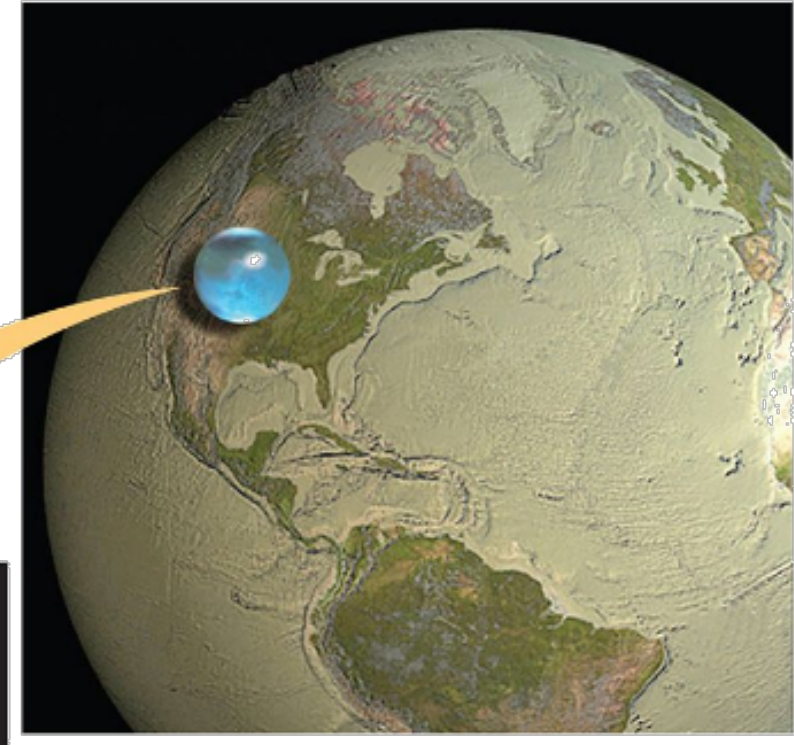
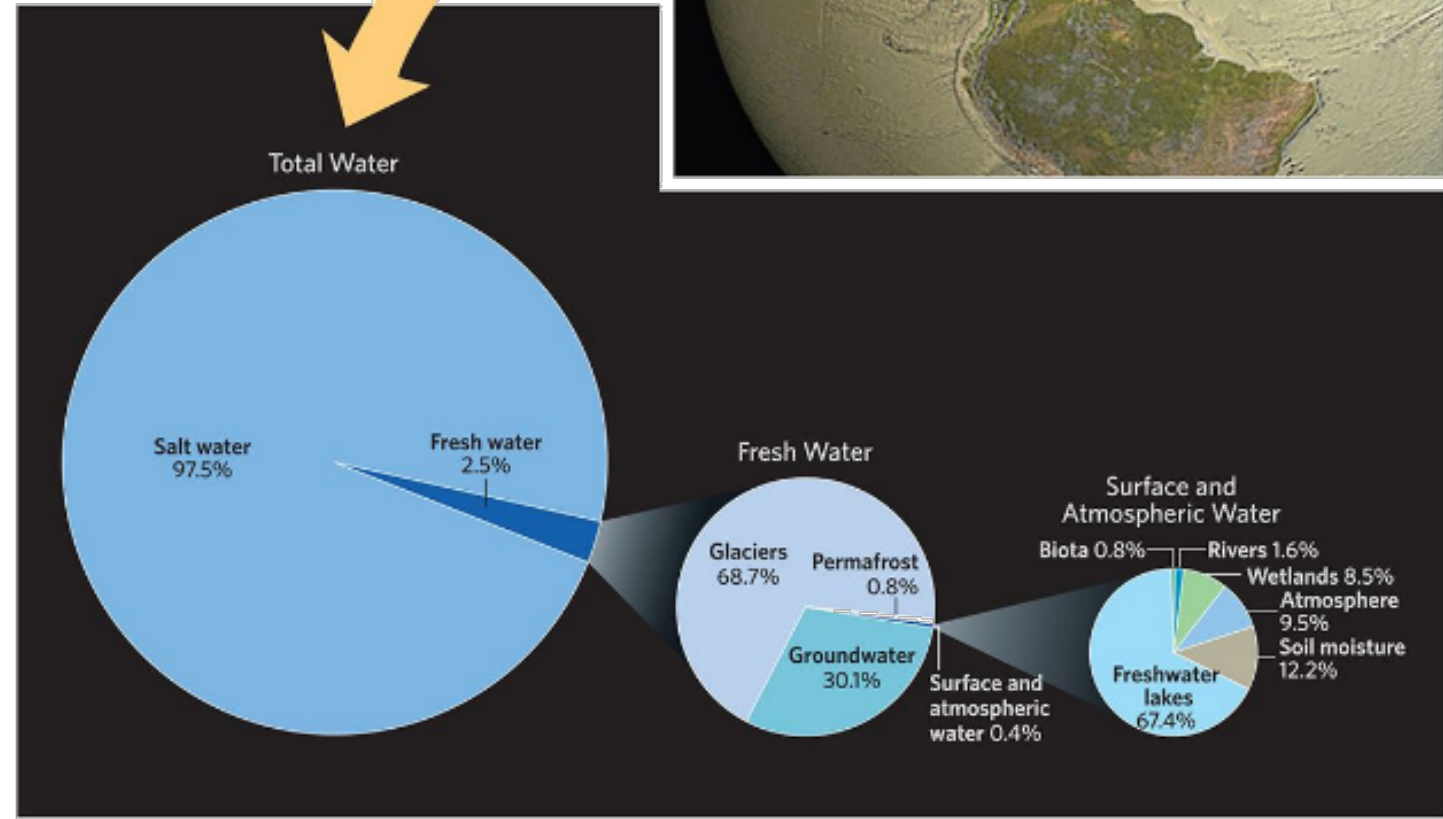
The Earth is an Ocean World

- Ocean definition: Vast body of saline water that occupies the depressions of the Earth's Surface
- The world ocean has a few natural divisions (Ex: continents), only one great interconnected mass of liquid water;
- Oceans are named for our convenience (Ex: Atlantic, Pacific, Indian, and Arctic Oceans;
- 71% of Earth's surface is covered by the world ocean

Getting some perspective:

Relative amount of water on the Earth's surface

More than 97% of the water lies in the ocean. If all water at Earth's surface were gathered up into a sphere, its diameter would measure only 1,380 km (860 miles)



Getting some perspective:

Relative amount of water on the Earth's surface

Those are huge numbers!

On a human scale, the ocean is mighty impressive



Statistics for the World Ocean

Total area:	331,441,932 square kilometers	127,970,445 square miles
Total volume:	1,303,155,354 cubic kilometers	312,643,596 cubic miles
Total mass:	1.41 billion billion metric tons	1.55 billion billion tons
Average depth:	3,682 meters	12,081 feet
Greatest depth:	10,994 meters	36,070 feet

Getting some perspective:

Relative amount of water on the Earth's surface

On a planetary scale, not so much

- The average depth is a tiny fraction of Earth's radius.
- Accounts for only slightly more than 0.02% of Earth's mass (0.13% of its volume).
- Much more water is trapped within the Earth's hot interior than exists in its ocean and atmosphere.

Ocean is over a
very thin crust

